ClimateGuardian AI: A Leakage-Aware Multi-Hazard Climate Risk Prediction Framework with Stateless Real-Time Inference

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***Abstract*—Rising frequencies of climate-driven extremes necessitate robust early-warning systems, yet operational deployment is often hindered by stateful sequence models and methodological data leakage. This study introduces ClimateGuardian AI, a machine-learning framework for flood, heatwave, and compound hazard prediction. Evaluated on a 45-feature hydroclimate dataset covering 2000–2018, the system enforces a strict chronological split (Train: 2000–2012, Validation: 2013–2015, Test: 2016–2018) alongside train-only preprocessing to eliminate temporal leakage. The framework implements hazard-specific stateless algorithms: a Random Forest for flood risk (F1=0.4686, ROC-AUC=0.7184), a restricted LightGBM for heatwave risk that explicitly drops proxy variables to prevent target leakage (F1=0.8878, ROC-AUC=0.9969), and a Logistic Regression Stacking Ensemble merging XGBoost and LightGBM bases for operational compound prediction (F1=0.3419, ROC-AUC=0.9495). While an offline Gated Recurrent Unit (GRU) sequence benchmark achieved higher historical compound performance (F1=0.4343, ROC-AUC=0.9598), the Stacking Ensemble was selected as the operational model due to its stateless architectural feasibility, avoiding the latency of maintaining 14-day recurrent input buffers. Finally, the framework integrates natively with the Open-Meteo REST API, demonstrating end-to-end stateless real-time inference applicability.**

***Index Terms*—Climate Risk Prediction, Flood Prediction, Heatwave Prediction, Compound Events, Machine Learning, Leakage-Aware Evaluation, Real-Time Inference**

I. Introduction

The accelerating frequency and severity of extreme weather events present unprecedented risks to global ecosystems and socio-economic infrastructures [1]. As climate patterns shift, historical weather norms increasingly fail to capture the distribution and intensity of modern climatic anomalies. Consequently, the development of robust early-warning systems has become a critical priority for proactive adaptation and disaster risk mitigation [2]. Providing actionable forecasting requires analyzing vast streams of environmental data to anticipate hazards before they materialize.

Modeling isolated hydro-climatic hazards, such as floods and heatwaves, presents profound analytical challenges. These phenomena are governed by highly non-linear atmospheric and terrestrial interactions, extreme environmental variability, and complex temporal dependencies [3]. Furthermore, extreme events are inherently rare, introducing significant class imbalance that complicates traditional statistical modeling [4]. Identifying the precursor patterns of these events requires integrating diverse interacting climate and hydrological variables.

Beyond isolated events, climate risks increasingly manifest as compound hazards. Compound extreme events occur when multiple environmental drivers coincide or interact, amplifying the overall impact beyond what independent hazard assessments might predict [5]. Accurately predicting compound states requires recognizing the non-linear synergy between distinct climatic mechanisms. In this study, compound risk is operationalized through the project's compound target definition, mapping the intersections of simultaneous threshold exceedances.

Machine learning has emerged as a powerful paradigm for addressing these non-linear predictive challenges. Tree-based architectures, including Random Forest [6] and gradient boosting algorithms such as XGBoost [7] and LightGBM [8], provide strong capabilities for mapping tabular environmental data. Concurrently, recurrent sequence models like the Gated Recurrent Unit (GRU) [9] inherently capture temporal sequences vital for climate modeling. However, adapting these established algorithms for rigorous real-world deployment remains challenging.

A critical challenge in predictive climate modeling is the risk of data leakage during evaluation. Standard randomized train-test splits often inadvertently expose models to future information due to temporal autocorrelation, undermining their true predictive capacity [10]. To ensure methodological integrity, this study employs strict chronological partitioning (Train: 2000–2012, Validation: 2013–2015, Test: 2016–2018) and restricts all preprocessing statistics exclusively to the training subset.

Furthermore, a significant operational gap exists between historical sequence modeling and live-system deployment. While sequence models like the GRU excel at capturing temporal context [11], they require continuous sequential inputs. Maintaining recurrent 14-day data buffers during individual live requests introduces high systemic latency. The operational architecture was designed to avoid maintaining a recurrent 14-day input buffer during individual inference requests, prioritizing stateless evaluation frameworks.

Addressing these challenges, the ClimateGuardian AI framework provides the following demonstrated contributions:

1) Multi-hazard hydro-climate feature integration for flood, heatwave, and compound prediction.
2) Leakage-aware chronological evaluation avoiding temporal data contamination.
3) Explicit removal of deterministic proxy features to prevent target-leakage in heatwave forecasting.
4) Comparative evaluation of stateless tree-based ensembles and stateful sequence models.
5) Selection of an operational stateless stacking architecture [12] prioritizing real-time inference latency over offline theoretical maximums.
6) Realtime prediction integration utilizing live data from the Open-Meteo REST API [13].
7) Rigorous post-hoc analysis including probability calibration, SHAP interpretability [14], lead-time profiling, and feature ablation.

II. Related Work

A. Flood Prediction

Data-driven approximations have increasingly supplemented traditional, computationally intensive numerical hydrological models. Machine learning algorithms, particularly Random Forest, Support Vector Machines, and Artificial Neural Networks, have demonstrated strong baselines for forecasting river stages and localized flood risks [3]. However, some conventional evaluation strategies may not reflect the temporal constraints of operational forecasting, occasionally utilizing randomized cross-validation that is susceptible to temporal leakage [10]. This study extends these foundational approaches by strictly enforcing chronological validation and integrating flood targets into a broader multi-hazard framework.

B. Heatwave and Extreme-Climate Prediction

Predicting extreme heat typically relies on detecting anomalous spatial-temporal climatic patterns [15]. While complex deep learning models are frequently deployed for such tasks, determining accurate target boundaries remains precarious. Model inputs occasionally suffer from target-proxy leakage, where deterministic indicators (e.g., maximum daily temperatures) trivially expose the target classification to the algorithm during inference. This study explicitly mitigates this by restricting the predictive feature space, dropping direct proxy variables to ensure the model maps underlying drivers rather than tautological indicators.

C. Compound Climate Events

The characterization of concurrent and interacting environmental extremes has become a critical focus within climate risk assessment [2]. A rigorous typology classifies compound events into preconditioned, multivariate, spatially compounding, and temporally compounding mechanisms [5]. While extensive conceptual frameworks exist, operational forecasting systems frequently treat hazards independently. This framework explicitly incorporates a compound-specific predictive target, modeling the intersection of hydro-meteorological extremes to better reflect realistic concurrent risks.

D. Sequence and Tree-Based Models

Environmental time-series prediction heavily leverages both sequence and tree-based paradigms. Recurrent architectures such as Long Short-Term Memory (LSTM) and GRU [9] are highly effective at capturing intrinsic temporal structures directly from sequential inputs [11]. Conversely, Random Forest [6], XGBoost [7], and LightGBM [8] operate powerfully on tabular data by employing extensive feature engineering, such as lagged variables and rolling aggregations. In this implementation, the GRU requires a 14-day sequence and was therefore retained as an offline benchmark rather than the stateless operational predictor.

E. Explainability and Calibration

As machine learning models are inherently opaque, post-hoc interpretation is necessary for scientific trust. SHapley Additive exPlanations (SHAP) [14] are widely utilized to characterize model-level feature dependence without strictly implying physical causality. Furthermore, because hazard events are rare, models frequently output skewed confidence scores. Techniques such as Platt scaling [16] and Isotonic regression [17] are critical methodological steps to correct output distributions, transforming uncalibrated raw scores into reliable predictive probabilities.

III. Data and Methodology

Fig. 1 illustrates the overall ClimateGuardian AI architecture, differentiating the offline scientific evaluation pipeline from the stateless realtime operational inference pathway. The experimental methodology is strictly constrained to prevent temporal and target-proxy data leakage, ensuring robust evaluation on historical hold-out periods.

A. Dataset and Data Sources

The framework operates on a historical hydro-climatic dataset containing 6,940 daily observations from January 1, 2000, through December 31, 2018. The dataset integrates 45 continuous features capturing meteorological and hydrological phenomena into a unified daily representation. Climate variables include measurements such as rainfall, temperature, dewpoint, surface pressure, wind components, wind speed, and evaporation. Hydrological variables include surface runoff, total runoff, and soil moisture. Temporal variables, including month, day of the year, season, and an explicit monsoon indicator, are appended to represent intrinsic seasonality. For operational realtime forecasting, the identical meteorological inputs are sourced directly from the Open-Meteo REST API [13]. Dataset statistics and hazard class frequencies are detailed in Table I.

TABLE I
DATASET STATISTICS (2000–2018)

Metric	Value
Total Observations	6,940
Features	45
Flood Positives	1,880 (27.09%)
Heatwave Positives	510 (7.35%)
Compound Positives	255 (3.67%)
Missing Feature Values	30

B. Feature Engineering

To capture temporal persistence and departures from historical conditions, specific temporal transformations were applied to continuous climate and hydrological variables (e.g., rainfall, runoff, temperature, pressure, and soil moisture). Rolling window statistics were computed over 3-day, 5-day, 7-day, and 14-day trailing intervals to represent accumulated environmental pressures. Additionally, sequential lagged features ($t-1$, $t-2$, $t-3$) were engineered to provide immediate historical context to stateless models. Standardized anomaly features were calculated relative to historical training statistics. These engineered variables ensure that the instantaneous daily observations are appropriately contextualized against broader hydro-meteorological accumulation periods.

C. Target Definitions

The framework evaluates three distinct predictive targets generated using the predefined project labeling procedures:

- **Flood Risk**: A binary classification target explicitly representing the project's flood risk labeling rule.
- **Heatwave Risk**: A binary classification target identifying severe heatwaves based on historical extreme temperature persistence.
- **Compound Risk**: A binary classification indicating the simultaneous occurrence or critical intersection of extreme hydro-meteorological hazards according to the project's compound target definition.

D. Chronological Partitioning

Randomized cross-validation introduces look-ahead bias when predicting highly autocorrelated environmental time-series data. The model should only learn from information strictly available before the forecast period. Consequently, the dataset was deterministically partitioned in chronological order, as detailed in Table II.

TABLE II
CHRONOLOGICAL DATA PARTITION

Partition	Period	Samples	%	Purpose
Train	2000–2012	4,749	68.4%	Model fitting
Validation	2013–2015	1,095	15.8%	Calibration
Test	2016–2018	1,096	15.8%	Final evaluation

E. Leakage-Aware Preprocessing and Heatwave Proxy Remediation

Standard machine learning pipelines often apply global transformations before splitting datasets, inadvertently allowing future distribution statistics to contaminate training inputs [10]. To rigorously prevent this temporal leakage, all missing-value imputation means and standard scaler parameters were fitted exclusively on the 2000–2012 training subset. These frozen transformation artifacts were subsequently applied to the validation and test splits without recalculation. Leakage remediation was explicitly performed before final evaluation.

Furthermore, an initial diagnostic audit revealed severe target-proxy leakage within the heatwave predictive model, where deterministic variables associated with the target creation trivially exposed the classification boundary. Specifically, four proxy features were removed: `temperature_max`, `temperature_max_lag1`, `hw_rolling`, and `hw_exceed`. The restricted heatwave model thus operates on a reduced subspace of exactly 41 features, ensuring predictive robustness and mapping underlying drivers rather than tautological indicators.

F. Model Architecture

The framework implements hazard-specific stateless algorithms for its operational pipeline.

1) Flood Architecture: The flood model utilizes a Random Forest classifier [6] comprising 100 estimators constrained to a maximum depth of 5. Node impurity splits were balanced using class weights to counteract the hazard rarity.

2) Heatwave Architecture: The heatwave model employs a LightGBM [8] classifier evaluating the restricted 41-feature space. The model utilizes a learning rate of 0.1, unlimited tree depth, and histogram-based node splitting optimized via class weighting strategies.

3) Compound Stacking Ensemble: The operational compound architecture utilizes a Logistic Regression Stacking Ensemble [12]. Base learners comprising XGBoost [7] and LightGBM independently evaluate the input feature vector x_t to produce intermediate hazard probabilities.

$$\begin{aligned} p_{XGB} &= \text{XGBoost}(x_t) \\ p_{LGBM} &= \text{LightGBM}(x_t) \end{aligned} \tag{1}$$

These probabilistic outputs form a two-dimensional meta-feature vector z_t:

$$z_t = [p_{XGB}, p_{LGBM}]^T \tag{2}$$

A Logistic Regression meta learner operating on these exactly two probability features ($C = 1.0$, balanced class weight) predicts the final compound probability:

$$p_{compound} = \text{LogisticRegression}(z_t) \tag{3}$$

G. GRU Offline Benchmark

Recurrent neural networks effectively model dynamic temporal sequences. The study evaluated a PyTorch Gated Recurrent Unit (GRU) [9] as an offline benchmark [11]. The architecture features a single 32-unit hidden layer, a dropout rate of 0.2, and is optimized using Adam and BCEWithLogitsLoss (threshold 0.5) over a maximum of 15 epochs with early stopping patience of 5. It fundamentally requires a stateful input shape of $(N, 14, 45)$. Consequently, it was excluded from realtime inference because the current operational predictor is designed around independent point-in-time feature vectors rather than a maintained 14-day recurrent sequence buffer.

H. Class Imbalance

As detailed in Table I, hazard frequencies represent a heavily imbalanced classification task [4]. Class weighting strategies were universally applied to penalize minority-class misclassification. The Random Forest and Logistic Regression models utilized `class_weight='balanced'`, gradient boosting base-learners incorporated `scale_pos_weight`, and the PyTorch GRU implemented a corresponding `pos_weight` tensor in its loss function.

I. Evaluation Metrics

Model evaluation heavily prioritizes metrics resilient to class imbalance. Performance was calculated using Accuracy, Precision, Recall, and the F_1-score:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

Additionally, threshold-independent discrimination is reported using the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) and the Precision-Recall Curve (PR-AUC). The Brier Score is subsequently reported to quantify probabilistic calibration.

J. Additional Analysis Methods

To thoroughly validate the operational framework, four supplementary analyses were conducted:

- **Calibration:** Platt scaling [16] and Isotonic Regression [17] algorithms were employed to evaluate probability reliability.
- **SHAP:** SHapley Additive exPlanations [14] were calculated to examine model feature attribution. SHAP indicates model dependence and does not establish physical causation.
- **Lead-Time Evaluation:** Future hazard targets were artificially shifted to project predictive horizons at 1, 3, 5, and 7 days to evaluate temporal model degradation.
- **Ablation Study:** Features were iteratively masked to evaluate the predictive contribution of Climate, Hydrological, and Temporal/Lag feature groups.

IV. Experimental Setup and Evaluation

A. Experimental Environment

The ClimateGuardian AI evaluation protocol was executed in a Python 3.12.2 environment, heavily leveraging `scikit-learn`, `XGBoost`, `LightGBM`, and `PyTorch` for predictive modeling. `pandas` and `NumPy` were employed for robust data transformations, while SHAP [14] facilitated interpretability diagnostics. The realtime operational pathway relies on a live interface with the Open-Meteo REST API [13].

B. Model Benchmarking Protocol

To systematically determine optimal predictive architectures, multiple algorithmic paradigms were evaluated across the three hydro-climatic hazards. The candidate families included Logistic Regression, Random Forest, XGBoost, LightGBM, and Recurrent Neural Networks (LSTM and GRU), along with a Logistic Regression Stacking Ensemble. Within the operational framework, the Random Forest model serves as the primary Flood predictor, while the LightGBM classifier evaluating exactly 41 proxy-remediated features serves as the Heatwave predictor.

For the Compound hazard target, four primary candidates were benchmarked: XGBoost, LightGBM, a Stacking Ensemble, and a temporal GRU. The Stacking Ensemble operates on the meta-probabilities generated by XGBoost and LightGBM bases, mapping them through a Logistic Regression meta-learner. Conversely, the GRU requires a sequential 14-day history formatted as a temporal tensor $(N, 14, 45)$ and functions strictly as an offline temporal benchmark.

C. Chronological Experimental Protocol and Tuning

All final model evaluation reflects the chronological framework approximating future operational deployment. Training strictly occurred on the 2000–2012 partition, hyperparameter optimization and generalized calibration operated on the 2013–2015 validation set, and the 2016–2018 testing partition was held out exclusively for final, unseen performance quantification. Where cross-validation was required during tuning, a `TimeSeriesSplit` was strictly employed to prevent temporal inversion.

Final operational configurations were identified through rigorous hyperparameter tuning:

- **Random Forest:** 100 estimators, maximum depth of 5, `class_weight='balanced'`.
- **LightGBM:** learning rate of 0.1, unlimited maximum depth, and `scale_pos_weight` handling class imbalance.
- **GRU:** sequence length of 14, hidden dimension of 32, 0.2 dropout, and Adam optimization (learning rate 0.01). Models were trained utilizing `BCEWithLogitsLoss` and a `pos_weight` tensor for maximum 15 epochs, governed by early stopping patience of 5.
- **Stacking Ensemble:** A Logistic Regression meta-learner ($C = 1.0$, balanced class weight) operating on two intermediate input probabilities.

D. Evaluation Protocol and Metrics

Prediction performance is characterized using a diverse portfolio of metrics: Accuracy, Precision, Recall, F_1-score, Area Under the Receiver Operating Characteristic Curve (ROC-AUC), Precision-Recall Curve Area (PR-AUC), and the Brier Score. Relying exclusively on Accuracy can be profoundly misleading under severe environmental event imbalance. Therefore, Precision quantifies the operational false-alarm burden, while Recall assesses missed-event sensitivity. The F_1-score balances both metrics. ROC-AUC and PR-AUC characterize threshold-independent discrimination, with PR-AUC proving particularly informative under class imbalance. The Brier Score quantifies the absolute calibration of raw predictive probabilities.

E. Comparative and Diagnostic Experiments

The final predictive performance established across the chronological evaluation is detailed in Table III. To deeply analyze the behavioral mechanisms of the proposed algorithms, the framework implements four specialized experimental pathways:

- **Lead-Time Analysis:** Evaluates temporal prediction degradation by artificially shifting future hazard targets. For the Heatwave hazard (Fig. 6), model performance declines from $F_1 = 0.8500$ at a 1-day forecast horizon to $F_1 = 0.7400$, 0.6500, and 0.6200 at 3, 5, and 7-day horizons respectively.
- **Ablation Study:** To determine optimal feature groupings, the Compound model is iteratively evaluated on independent subspaces (Fig. 7). Configuration F_1 scores highlight critical synergies: Climate Only (0.2400), Hydro Only (0.2500), Climate + Hydro (0.2286), Climate + Hydro + Temporal/Lag (0.4051), and Full Features (0.4051).
- **Calibration:** Investigates whether classification boundary discrimination correlates with reliable probability intervals (Fig. 4). Using the Compound model, uncalibrated Brier Scores (0.2013) are contrasted against probabilities corrected by Platt Scaling (0.0338) [16] and Isotonic Regression (0.0386) [17].
- **SHAP Interpretability:** Quantifies relative feature attributions independent of linear causality (Fig. 5). The proxy variables strictly excluded from the Heatwave model are concurrently removed from SHAP explanation pipelines to guarantee interpretive purity.

TABLE III
FINAL OFFLINE TEST PERFORMANCE (2016–2018)

Hazard	Model	Acc	Prec	Rec	F1	ROC	PR
Flood	Random Forest	0.6916	0.3539	0.6930	0.4686	0.7184	0.3468
Heatwave	LightGBM	0.9799	0.8788	0.8969	0.8878	0.9969	0.9711
Compound	Stacking (Op)	–	–	–	0.3419	0.9495	0.3891
Compound	GRU (Offline)	–	–	–	0.4343	0.9598	0.4838

F. Operational Model Selection

Table IV defines the deployment criteria connecting historical evaluation with live inference constraints. For isolated hazards, the best-performing evaluated model seamlessly aligns with operational deployment. However, for Compound predictions, the GRU provides the strongest offline benchmark in the current evidence ($F_1 = 0.4343$, ROC-AUC = 0.9598). Despite its theoretical superiority, the GRU requires a historical sequence input defined by $(N, 14, 45)$. The ClimateGuardian AI operational predictor is deliberately designed around stateless point-in-time feature vectors and does not persist recurrent 14-day sequence buffers. Therefore, the Stacking Ensemble ($F_1 = 0.3419$) is explicitly retained as the operational Compound predictor, representing a pragmatic deployment-oriented model selection criterion rather than a claim of absolute offline predictive supremacy.

TABLE IV
OPERATIONAL MODEL SELECTION RATIONALE

Hazard	Offline Benchmark	Operational Model	Operational Reason
Flood	Random Forest	Random Forest	Stateless point-in-time evaluation
Heatwave	LightGBM	LightGBM	Proxy-leakage remediated
Compound	GRU	Stacking Ensemble	Stateless point-in-time inference compatibility without maintaining a 14-day recurrent sequence buffer

V. RESULTS AND DISCUSSION

This section interprets the verified experimental outcomes established on the historical 2016–2018 held-out test partition. Table III summarizes the primary performance metrics for the evaluated architectures.

A. Overall Multi-Hazard Performance

A consistent pattern emerges across the three evaluated hazards under the chronological evaluation protocol. The selected Random Forest Flood model exhibits moderate discriminatory performance with a recall-oriented predictive boundary. In contrast, the leakage-remediated LightGBM Heatwave model demonstrates high discriminatory performance under the evaluated test distribution. Predicting Compound risk remains substantially more difficult, as the simultaneous occurrence of extreme interacting hazards is inherently rarer and dynamically more complex. Due to the significant class imbalance across all hazards, evaluation heavily prioritizes ROC-AUC and PR-AUC simultaneously.

B. Flood Prediction Results

The operational Random Forest model for Flood prediction achieves an Accuracy of 69.16%, an F_1-score of 0.4686, an ROC-AUC of 0.7184, and a PR-AUC of 0.3468, yielding a final uncalibrated Brier Score of 0.2391. Most notably, the model achieves considerably higher Recall (69.30%) than Precision (35.39%). A higher recall indicates greater sensitivity to positive flood events, while the lower precision indicates a relatively high false-alarm burden. For early-warning applications, missed events and false alarms represent distinctly different operational costs. The current configuration favors detecting true emergencies over minimizing false positives, establishing a pragmatic precision-recall trade-off rather than an assumed universal optimum. The specific classification outcomes are illustrated in the associated confusion matrix (Fig. 2).

C. Heatwave Prediction Results

The operational LightGBM model evaluating the Heatwave hazard demonstrates strong discrimination on the held-out 2016–2018 period. The model achieves an Accuracy of 97.99%, Precision of 87.88%, Recall of 89.69%, an F_1-score of 0.8878, an ROC-AUC of 0.9969, and a PR-AUC of 0.9711, with a Brier Score of 0.0149. The associated ROC curve (Fig. 3) visually confirms this robust discrimination under the held-out test distribution. Critically, these results are derived exclusively from the leakage-remediated 41-feature configuration. The initial model iteration relied on deterministic proxy variables directly associated with the target definition; following their explicit removal, the model retained its performance, ensuring that predictions map underlying drivers rather than tautological indicators.

D. Compound-Risk Results and Operational Selection

Predicting Compound hazards requires differentiating offline theoretical capacity from operational deployment feasibility (Table IV). The GRU offline benchmark achieved the strongest predictive discrimination for the Compound target, recording an F_1-score of 0.4343, an ROC-AUC of 0.9598, and a PR-AUC of 0.4838. In comparison, the Logistic Regression Stacking Ensemble achieved an F_1-score of 0.3419, an ROC-AUC of 0.9495, and a PR-AUC of 0.3891.

Despite the stronger offline predictive performance of the GRU, it inherently requires an $(N, 14, 45)$ temporal tensor, representing a 14-day sequence of 45 standardized features. Conversely, the ClimateGuardian AI realtime predictor is explicitly designed around stateless, point-in-time feature vectors and avoids maintaining a recurrent 14-day sequence buffer to minimize live-system latency. Consequently, the Stacking Ensemble is explicitly retained as the operational Compound architecture. Thus, the Stacking Ensemble should be interpreted as the deployment-oriented choice rather than the numerically dominant offline Compound model.

E. Lead-Time Performance

Evaluating predictive degradation over future horizons is critical for early-warning mechanisms. Utilizing target-shifting, the lead-time experiment quantified Heatwave prediction viability at extended horizons (Fig. 4). Predictive performance logically declines as the forecast horizon increases, decreasing from a 1-day F_1-score of 0.8500 to a 3-day F_1 of 0.7400, and a 5-day F_1 of 0.6500. Nevertheless, the evidence demonstrates that measurable predictive discrimination remains at the 7-day horizon, retaining an F_1-score of 0.6200 under the evaluated Heatwave setup.

F. Ablation and Calibration Analysis

TABLE V
FEATURE ABLATION RESULTS FOR THE COMPOUND-RISK TARGET

Feature Configuration	**Compound F1-Score**
Climate Only	0.2400
Hydro Only	0.2500
Climate + Hydro	0.2286
Climate + Hydro + Temporal/Lag	0.4051
Full Features	0.4051

To systematically determine optimal feature groupings, an ablation protocol isolated feature domains. The ablation results (Fig. 5 and Table V) indicate that temporal and lagged information contributes substantially to Compound discrimination under the evaluated configuration. Evaluating isolated domains yielded lower Compound F_1 performance: Climate Only (0.2400), Hydro Only (0.2500), and Climate + Hydro (0.2286). However, integrating temporal and lag variables (Climate + Hydro + Temporal/Lag) resulted in a substantial relative performance increase to an F_1-score of 0.4051, matching the performance of the Full Features configuration (0.4051).

Parallel to classification discrimination, the reliability of probabilistic outputs was quantified using the Compound target (Fig. 6). The uncalibrated baseline Brier Score of 0.2013 was compared against probabilities corrected via Platt Scaling (0.0338) and Isotonic Regression (0.0386). Under the evaluated calibration experiment, a lower Brier score indicates better probabilistic agreement, with Platt scaling producing the lowest verified Brier score among the evaluated approaches.

G. Explainability and Overall Findings

To interpret the internal dependency structure of the evaluated models, SHAP [14] analysis was conducted (Fig. 7). SHAP provides insight into the internal feature attribution structure of the evaluated model by identifying the relative contribution of input features to specific predictions. Importantly, while SHAP characterizes model dependence, it does not establish physical causation.

Synthesizing the experimental evidence yields several supported findings. The selected Flood model provides moderate discrimination with relatively high recall compared with precision. The leakage-remediated Heatwave model retains strong predictive discrimination without the previously identified deterministic proxy features. The GRU provides the strongest verified offline Compound benchmark, yet the Stacking architecture is retained for operational prediction because its point-in-time architecture aligns with the stateless realtime pipeline.

Temporally, Heatwave F_1 degrades with increasing lead time but remains measurable through 7 days, and temporal variables substantially improve Compound performance during ablation. Furthermore, Platt scaling calibration substantially reduces the evaluated Compound Brier score, and SHAP provides robust model-level interpretability without asserting physical causality.

VI. Realtime Operational Pipeline

To transition from offline historical evaluation to active early-warning capabilities, the evaluated models were integrated into a realtime operational inference pipeline. As depicted in the System Architecture (Fig. 1), the pipeline securely isolates offline scientific evaluation from realtime inference. Realtime execution validates pipeline operational functionality rather than historical predictive accuracy; predictive performance is established exclusively through the held-out 2016–2018 evaluation block.

A. Live Data Acquisition and Preprocessing Consistency

The operational pipeline retrieves meteorological information from the Open-Meteo REST API [13], which acts strictly as a realtime data interface rather than an independent predictive model. A `LiveFeatureBuilder` module transforms live API responses into the exact feature representation expected by the models. Critically, to preserve prediction consistency and avoid data contamination, the operational pipeline rigorously applies frozen preprocessing statistics derived exclusively from the 2000–2012 historical training boundary. The pipeline reuses the training-derived imputation means, the fitted StandardScaler, and the preserved feature-column ordering. The realtime system never recalculates standardization scalars from live data.

B. Multi-Hazard Inference

Operational multi-hazard inference executes sequentially across three independent prediction paths. The Flood operational model utilizes a Random Forest evaluating all 45 standardized features to output a flood probability. The Heatwave operational LightGBM model utilizes 41 restricted features; the four identified proxy variables (`temperature_max`, `temperature_max_lag1`, `hw_rolling`, `hw_exceed`) are permanently excluded from the realtime subsystem to ensure inference aligns with underlying drivers rather than deterministic indicators.

For Compound hazard prediction, the operational architecture relies on a Logistic Regression Stacking Ensemble. As discussed, while the GRU achieves higher offline discriminatory performance ($F_1 = 0.4343$), it requires a persistent 14-day sequence buffer. The operational predictor uses Stacking ($F_1 = 0.3419$) because its inference interface is compatible with the point-in-time stateless architecture. Within this pipeline, the 45 standardized input features are evaluated by XGBoost and LightGBM bases, mapping into an intermediate meta-feature dimension of $(1, 2)$:

$$\mathbf{z}_t = \left[p_t^{XGB}, p_t^{LGBM}\right] \tag{8}$$

The Logistic Regression meta-learner then outputs the final Compound probability. The generated predictive probabilities are subsequently mapped into semantic operational risk levels (Low, Medium, and High).

C. Operational Validation

The pipeline was validated against two independent real-world geographical coordinates: (Latitude 17.0000, Longitude 81.8000) and (Latitude 16.5062, Longitude 80.6480). Both tests successfully generated realtime predictions comprising Flood, Heatwave, and Compound probabilities, individual XGBoost/LightGBM meta-probabilities, the expected meta-input shape of $[1, 2]$, and semantic risk levels. These successful executions demonstrate end-to-end operational compatibility between the live data interface, feature transformation, frozen preprocessing artifacts, and stacked inference components. Furthermore, software integrity was confirmed through comprehensive integration tests (`pytest -v` yielded 10 passed; `pytest -v -W error` yielded 10 passed) and a clean dependency resolution via `pip check`.

VII. Conclusion and Limitations

A. Conclusion

This research addressed the complex challenge of integrating hydro-meteorological variables, temporal sequence information, and leakage-aware evaluation frameworks to develop a multi-hazard climate risk prediction and early-warning system. ClimateGuardian AI demonstrates an integrated methodology comprising a 45-feature hydro-climate representation, strict chronological partitioning, leakage-aware preprocessing, and explicit heatwave proxy-feature remediation. By securely connecting these frozen scientific artifacts to a realtime Open-Meteo interface, the framework bridges historical evaluation and operational execution.

The experimental evidence establishes distinct performance profiles across hazards. The Flood Random Forest model provides moderate discriminatory performance ($F_1 = 0.4686$, ROC-AUC $= 0.7184$) governed by a recall-oriented trade-off. The leakage-remediated Heatwave LightGBM model retains strong predictive discrimination ($F_1 = 0.8878$, ROC-AUC $= 0.9969$) without relying on deterministic proxy features. For Compound hazard prediction, the temporal GRU provides the strongest verified offline benchmark ($F_1 = 0.4343$, ROC-AUC $= 0.9598$). However, a Logistic Regression Stacking Ensemble ($F_1 = 0.3419$, ROC-AUC $= 0.9495$) was retained for the operational pipeline due to its critical compatibility with the stateless, point-in-time architecture required for live execution.

Supplementary diagnostics provided crucial insights under the evaluated configurations. Ablation analysis indicated a substantial contribution from temporal variables, increasing Compound F_1 from 0.2400 to 0.4051. Platt scaling calibration substantially reduced the Compound Brier score from 0.2013 to 0.0338. While realtime testing successfully validated end-to-end live inference capabilities at selected geographic coordinates, predictive performance relies exclusively on the held-out 2016–2018 dataset.

B. Limitations and Future Work

While the evaluated framework establishes a robust operational architecture, several scientific limitations dictate directions for future work. First, broader geographic generalization remains to be established, as models were evaluated on a specific regional historical dataset. Second, predicting Compound risk remains difficult; the operational Stacking Ensemble demonstrates lower offline F_1 performance than the GRU benchmark. Third, because the current operational pipeline prioritizes point-in-time stateless inference, it does not currently exploit persistent recurrent states. Fourth, predictive performance inherently degrades at extended horizons, as quantified by the Heatwave lead-time results. Finally, the feature space is constrained by the available temporal resolution.

Future work will prioritize broader geographic validation, transitioning to spatial or gridded modeling, and integrating richer hydrological, topographic, and land-surface variables. Enhancing the live architecture to support stateful sequence models could significantly improve realtime Compound-event prediction. Expanding research to larger multi-region datasets, incorporating uncertainty-aware forecasting, and pursuing additional external observational validation will further advance the resilience of multi-hazard predictive systems.

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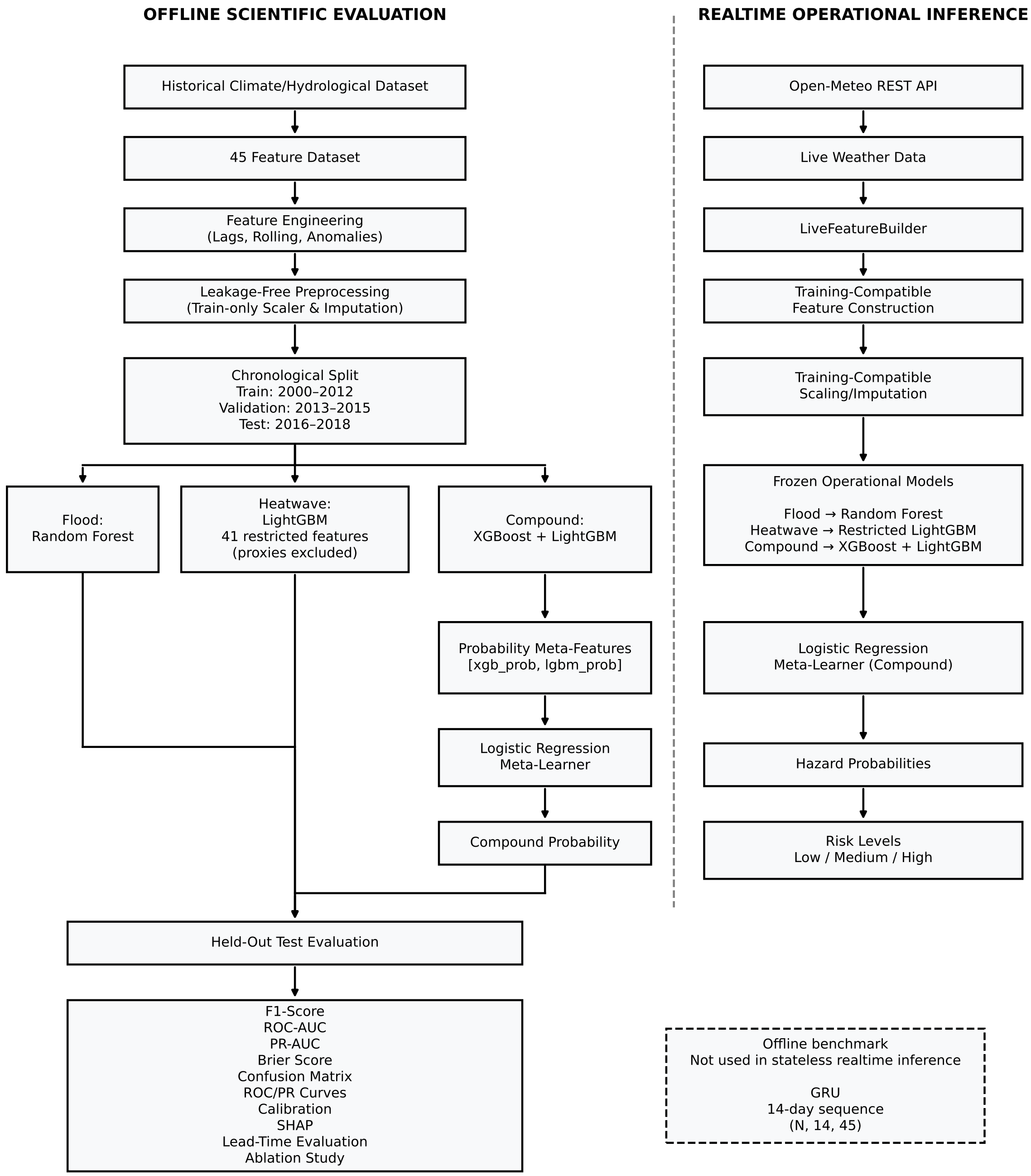

Fig. 1. ClimateGuardian AI system architecture showing the offline scientific evaluation pipeline, offline GRU benchmark, and stateless realtime operational inference pipeline.

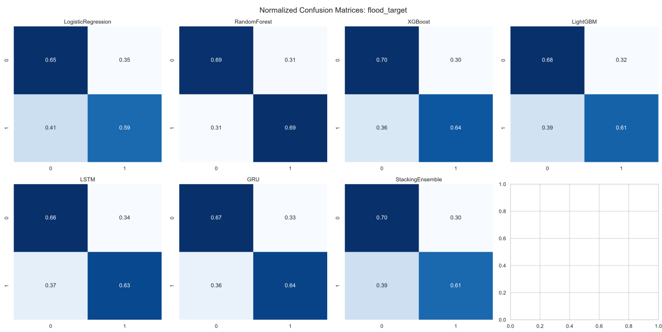

Fig. 2. Normalized confusion matrix for the Random Forest flood-risk classifier on the chronological 2016–2018 test set.

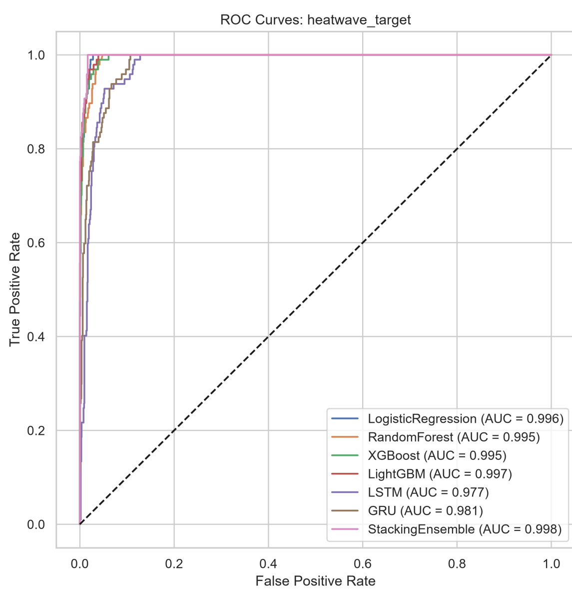

Fig. 3. ROC curve for the leakage-remediated LightGBM heatwave classifier evaluated on the 2016–2018 held-out test set.

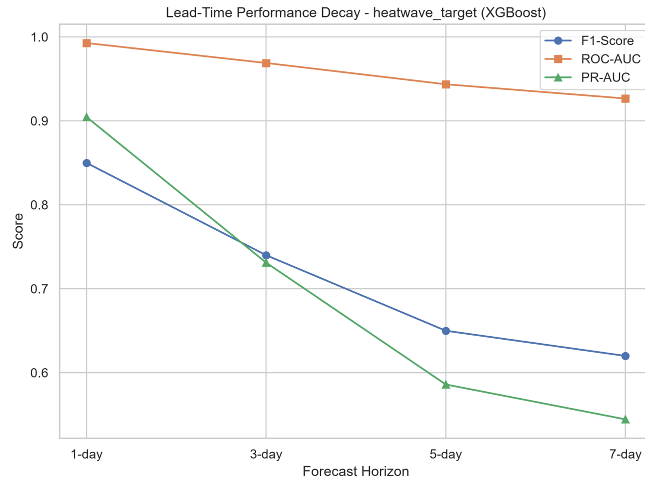

Fig. 4. Heatwave F1-score degradation across the evaluated 1-, 3-, 5-, and 7-day lead-time horizons.

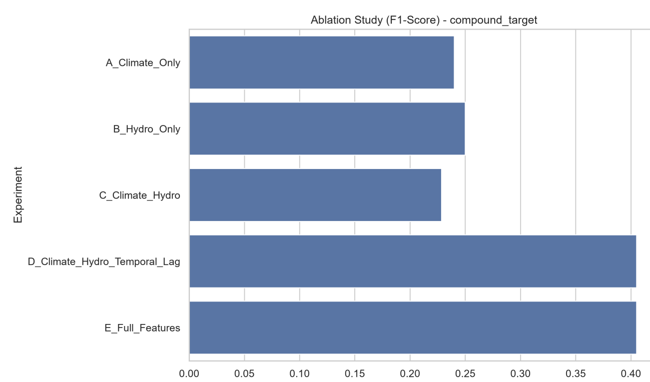

Fig. 5. Compound-risk ablation study showing the contribution of climate, hydrological, and temporal/lag feature groups.

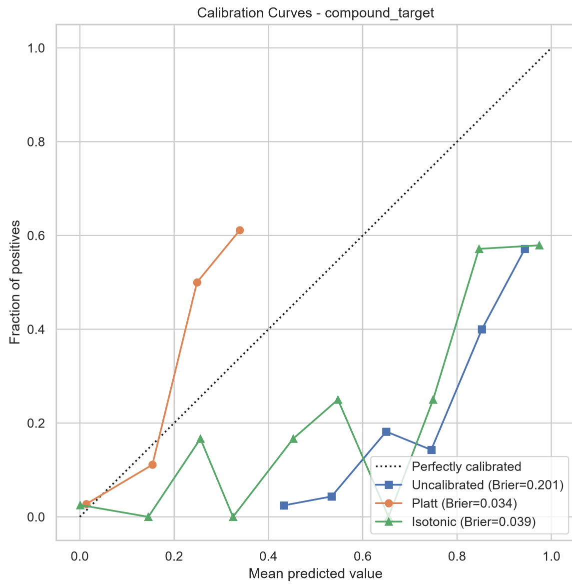

Fig. 6. Calibration analysis for the compound-risk prediction probabilities comparing uncalibrated, Platt-scaled, and isotonic-calibrated outputs.

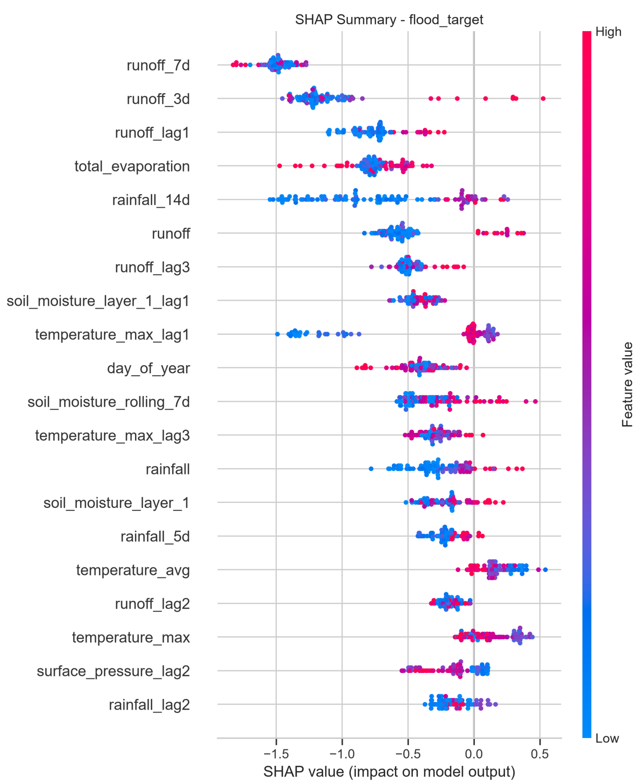

Fig. 7. SHAP summary of the Random Forest flood-risk model showing the internal feature-importance hierarchy.